

MEMORY DIVERGENCE IS NOT BEHAVIORAL DIVERGENCE: STAGE-RESOLVED TRANSPORT IN SELF-IMPROVING AGENT MEMORY

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ABSTRACT

Self-improving agents increasingly convert terminal feedback into persistent memory, but common evaluations conflate four distinct events: a feedback intervention changes stored state, that state is retrieved, the policy uses the retrieved difference, and behavior changes. We study this transport problem with frozen success-versus-failure writer interventions on otherwise identical trajectories. Reward-conditioned writing produces different persistent memories for all 20 Shopping pairs and all 4 Reddit pairs, and a stronger same-mode wording control leaves a significant reward-mode excess. When branch-specific memory is mechanically supplied, it has measurable downstream leverage (terminal $|\Delta| = 0.15625$, $p = 0.00074$). Under native reuse, however, Shopping still retrieves the branch-conditioned source-memory item in 125/172 opportunities while the first-action branch contrast is not supported (TV = 0.06944, $p = 0.5801$, 0/36 modal changes) and terminal transport is sparse ($|\Delta| = 0.02083$, $p = 0.4289$, 34/36 zero). Reddit repeats the write-stage effect but has sparse, sign-heterogeneous native terminal differences. Because write distance, retrieval rate, action TV, and endpoint differences are not commensurate, collapsing evaluation to any one surface creates observational aliases: write-only evaluation suggests a strong update, retrieval-only evaluation suggests active reuse, native endpoint-only evaluation suggests almost no effect, and forced-only evaluation suggests substantial behavioral leverage. We therefore introduce an ordinal stage-evidence ladder that preserves each stage’s own evidence semantics and localizes the first unsupported measured native stage after supported/observed prerequisites. The full stage signature reconciles these apparently conflicting summaries: on the frozen Shopping tests, action uptake is the first stage whose branch contrast is not supported after write and item-level exposure have been established. Because stage units and test sensitivities are not matched, this is an evidence-localization result, not an identified latent attenuation onset or causal mediation estimate. The result establishes a measurement principle for persistent self-improvement: state divergence and memory availability do not by themselves establish stable decision-level uptake.

1 INTRODUCTION

A persistent-memory agent can disagree with itself long before task accuracy reveals anything. Consider two copies of the same successful interaction trajectory. If one copy is labeled successful and the other failed before reflection, an outcome-conditioned memory writer may store different reusable advice even though the underlying experience is unchanged. That is already a meaningful state intervention. But it leaves a harder question unanswered: *where, if anywhere, does the resulting state difference become a behavioral difference?*

This distinction matters because persistent self-improvement contains several causal surfaces that are often collapsed into one “memory helped” statement. A terminal signal may first alter what is **written**; the resulting branch-conditioned source-memory item may or may not be **exposed** by retrieval on a later task; the policy may or may not **use** the treatment-specific information within that retrieved item; and only then can behavior change at the terminal **outcome**. A positive memory-similarity result identifies the first surface, while a retrieval hit only identifies availability. Neither is evidence

by itself that the agent’s decision rule changed. Conversely, an endpoint null does not reveal whether the update failed to write, failed to retrieve, or was filtered after retrieval. We call this *evaluation coarsening*: observing only a projection of the stage chain aliases distinct internal explanations into the same coarse success or failure label.

We study this problem through a stage-resolved intervention rather than a new memory algorithm. For a frozen trajectory τ , we switch the released success/failure reflection branch and construct paired persistent memories while holding the source experience fixed. This is a bundled writer-protocol intervention: the branch changes the reflection instruction together with its outcome/reward semantics, so it is not an atom-level isolation of a reward bit. We then instrument four native stages,

write $\rightarrow$ retrieval exposure $\rightarrow$ first-action uptake $\rightarrow$ terminal outcome,

and add a separate forced fixed-evidence control that bypasses native retrieval. The forced arm is deliberately *not* another stage in the chain: it asks whether branch-specific memory content has downstream leverage when supplied, whereas the native path asks whether the deployed memory system actually transports that difference into behavior.

This design distinguishes several explanations that an endpoint-only experiment cannot separate. If the writer does not create a persistent intervention, the story stops at write time. If the downstream policy is globally insensitive to memory content, even forced exposure should have little leverage. If the only problem is retrieval absence, native exposure should collapse. If retrieval itself is a valid proxy for policy use, observed exposure should co-occur with a supported action-level branch contrast. Finally, if reward-conditioned memory has a universal directional effect, terminal differences should not be dominated by zeros and sign reversals across tasks. Because these stages have different units, sample units, tests, and experimental surfaces, we do not convert them into a single “transport efficiency.” Instead, we retain a *stage signature* and introduce an ordinal stage-evidence ladder: preserve each stage’s own inferential/descriptive semantics, then localize the first *measured* native stage whose branch contrast is no longer supported after the preceding prerequisites have been supported or directly observed. The stage tests are not power-matched, so this operator localizes an evidence boundary rather than the latent point where a causal effect must physically decay. The point is diagnostic disambiguation rather than a finer-grained score.

The frozen evidence yields a sharp pattern. Reward-conditioned writing changes durable memory in all 20 Shopping pairs and all 4 Reddit pairs. A same-mode wording control shows that the reward/reflection-mode contrast exceeds a stronger within-mode wording perturbation. Forced fixed-evidence exposure produces terminal $|\Delta| = 0.15625$ ($p = 0.00074$), demonstrating downstream leverage under supplied memory. Yet native Shopping retrieval remains common—125 hits in 172 held-out opportunities—while first-action TV is only 0.06944 ($p = 0.5801$) with 0/36 modal action changes, and native terminal $|\Delta|$ is 0.02083 ($p = 0.4289$) with 34/36 zero cells. Reddit again shows reliable write divergence but sparse terminal transport: 6/8 cells are zero and the two nonzero effects have opposite signs.

Taken together, these measurements weaken three simple explanations at once: the native null is not caused by failure to create persistent state; it is not well described as global downstream insensitivity to supplied memory content; and it is not explained by retrieval absence alone. More importantly, they expose why coarse evaluations disagree. Write-only evaluation would call the update strong; retrieval-only evaluation would call reuse active; native endpoint-only evaluation would call the effect negligible; forced-only evaluation would call the memory behaviorally consequential. The full stage signature resolves this apparent contradiction: on the frozen Shopping tests, first-action uptake is the first measured native stage whose branch contrast is not supported after durable write and source-item exposure have been established. A replay-qualified post-hoc collision analysis of the historical repeated first-action decodes also does not support aggregate success/failure separation after correcting for within-condition stochastic concentration. We intentionally stop short of a causal mediation claim because the writer intervention remains bundled and lacks per-atom decoding-seed binding, the collision reanalysis is not an independent prospective replicate, and the forced and native surfaces are not identical interventions.

This diagnosis leaves a narrower prospective control question. Native source-item exposure is common (125/172 opportunities), yet a stable branch-conditioned first-action distribution shift is not supported by the frozen test. The completed experiment therefore establishes an *operational separation between availability and decision-level uptake*; it does not directly observe latent state relevance,

latent “behavioral authority,” or the mechanism that caused attenuation after exposure. One testable successor hypothesis is that post-exposure authorization should depend on reproducible matched-state branch sensitivity rather than relevance alone. We freeze one such selective-authority protocol prospectively, but move its controller details to the Discussion/Appendix because no downstream method-effect experiment has completed.

Our contributions are:

1. **A same-trajectory stage-resolved intervention for feedback-conditioned memory transport.** Rather than claiming that retrieval-versus-reuse separation is itself new, we hold the source trajectory fixed, switch the released success/failure reflection branch as a bundled writer-protocol intervention, and follow the resulting persistent-state contrast through native exposure, first-action uptake, and terminal outcome.
2. **A capacity-versus-transport control.** Forced fixed-evidence exposure establishes that branch-specific memory can have downstream leverage, while native reuse measures whether the deployed retrieval-and-policy path realizes that leverage.
3. **A diagnostic stage signature that exposes evaluation coarsening.** We show that write-only, retrieval-only, native-endpoint-only, and forced-only views of the same frozen system support different coarse summaries. We preserve stage-specific evidence states (supported, directly observed, or not supported) and define an evidence-localization boundary as the first unsupported measured native stage after supported/observed prerequisites, without inventing a cross-stage ratio, power equivalence, or causal mediation coefficient.
4. **A cross-domain boundary rather than a universal effect claim.** Reddit reproduces write-stage divergence but shows sparse and sign-heterogeneous native terminal effects, ruling out a simple monotone “more memory difference means more behavioral difference” story on the frozen evidence.

The resulting frozen claim remains narrower than a demonstrated new memory method: **persistent-state divergence and retrieval availability do not by themselves establish stable decision-level uptake.** The boundary supplies a concrete next causal question—whether selective post-exposure authorization can improve transport relative to an equally sparse random authorization rule—but that question is prospective. The present paper is therefore a stage-resolved measurement/identification study, not an evaluated memory-arbitration method paper.

2 STAGE-RESOLVED MEMORY TRANSPORT

2.1 CONTROLLED WRITE INTERVENTION

Let τ denote a frozen interaction trajectory before persistent memory construction. We create two writer conditions on the same τ ,

$$r \in \{S, F\}, \quad m^{(r)} = g(\tau, r),$$

where g is the released memory writer and S/F selects its success- or failure-reflection mode. The operational write contrast is

$$W(\tau) = d_{\text{mem}}\left(m^{(S)}, m^{(F)}\right).$$

$W > 0$ establishes that the frozen writer produced different persistent states. It does *not* establish that every textual atom is a pure causal effect of the reward bit, because decoding randomness is not separately noise-calibrated at the atom level. We therefore use the same-mode wording control in Section 4 to test the stronger alternative that ordinary prompt wording alone explains the observed write divergence.

2.2 NATIVE TRANSPORT IS A SEQUENCE OF DISTINCT OBSERVABLES

For a held-out future query q , define native retrieval exposure at the source-item level,

$$E_r(q) = \mathbf{1}[m^{(r)} \in \text{Retrieve}(q; \mathcal{M}^{(r)})].$$

Exposure is an availability variable: it says that the branch-conditioned source-memory item reached the policy context. It does not identify which treatment-specific content inside that item was read or used by the policy.

We therefore measure the first downstream decision boundary separately. Let $p_r^{(1)}(\cdot | q)$ be the first-action distribution under branch r . The action-level contrast is

$$U(q) = \text{TV}\left(p_S^{(1)}(\cdot | q), p_F^{(1)}(\cdot | q)\right).$$

Finally, for terminal outcome $y_r(q)$,

$$O(q) = |y_S(q) - y_F(q)|.$$

The native scientific object is therefore the ordered observation tuple

$$\mathcal{T}_{\text{native}}(q) = (W, E, U, O),$$

not any one component in isolation.

Crucially, these components are **not commensurate**. W is a memory-state distance, E is an exposure indicator/rate, U is a distributional action distance, and O is an endpoint contrast. We therefore do not divide them into a scalar “transport efficiency” or treat their numerical differences as a mediation decomposition. The order is mechanistic; the scales are not.

A further identifiability issue is *evaluation coarsening*. Let $\Pi_A(\mathcal{T}_{\text{native}})$ project the chain onto observed stages $A \subseteq \{W, E, U, O\}$. The projection is many-to-one: for example, a small endpoint under $\Pi_{\{O\}}$ is compatible with writer failure, retrieval absence, post-retrieval filtering, or sparse heterogeneous transport, while $\Pi_{\{E\}}$ cannot distinguish availability from policy uptake. We therefore retain the categorical stage signature $\Sigma = (z_W, z_E, z_U, z_O; z_F)$, with z_F denoting the forced-capacity side control. It is not a scalar metric; it preserves which explanations remain distinguishable after each stage is observed.

2.3 FORCED LEVERAGE IS A CAPACITY CONTROL, NOT A NATIVE STAGE

The forced fixed-evidence experiment mechanically supplies paired memory content to the downstream task packet. Let

$$L_{\text{forced}}(q) = |y^{\text{forced},S}(q) - y^{\text{forced},F}(q)|.$$

Because native retrieval is bypassed, L_{forced} is not inserted between W and E . It answers a different question: *can* branch-specific memory content affect the downstream system when supplied? Native (E, U, O) asks whether the deployed memory path *does* transport that contrast. This capacity-versus-transport separation is the key control that prevents a weak native endpoint from being summarized as “memory can never matter.”

2.4 AN ORDINAL STAGE-EVIDENCE LADDER

The central measurement problem is that W , E , U , and O cannot be compared on one numerical scale, and their frozen tests do not have matched sample units or statistical power. We therefore localize an *evidence boundary* with an ordinal ladder rather than a post-hoc transport coefficient or a latent bottleneck estimator. Let the ordered native stages be

$$s_1 = \text{write}, \quad s_2 = \text{retrieval exposure}, \quad s_3 = \text{first-action uptake}, \quad s_4 = \text{terminal outcome}.$$

For each stage we preserve its native evidence semantics. A stage may be SUPPORTED by its frozen inferential test, OBSERVED directly when the claim is descriptive rather than thresholded, or NOT-SUPPORTED when its frozen primary test does not establish a stable branch contrast. We then define

$$b^* = \min\{j : s_j \text{ is not supported/observed while } s_1, \dots, s_{j-1} \text{ are supported/observed}\}.$$

This is an ordering rule, not a ratio: no cross-stage units are divided, normalized, or interpreted as mediation mass. Because the stage tests are not power-matched, b^* means “first unsupported measured stage under the frozen tests,” not “the stage where the true causal effect first becomes small.”

The forced fixed-evidence arm is evaluated separately as a *side capacity condition*. It can weaken the global-insensitivity explanation, but because it bypasses native retrieval it cannot make an otherwise unsupported native stage pass. This prevents the positive forced endpoint from being inserted into the native chain merely because it is statistically strong.

The same ladder can be read as an alternative-explanation audit:

A1: no state intervention $\Rightarrow W \approx 0$,

A2: global downstream insensitivity $\Rightarrow L_{\text{forced}} \approx 0$,

A3: retrieval absence only $\Rightarrow E \approx 0$,

A4: exposure is uptake $\Rightarrow$ observed E should be accompanied by supported U ,

A5: universal directional transport $\Rightarrow O$ should be broadly nonzero with stable sign.

In our frozen evidence, write is SUPPORTED, native source-item exposure is OBSERVED, first-action uptake is NOT-SUPPORTED, and terminal transport is sparse and sign-heterogeneous. Hence $b^* = s_3$: **first-action uptake is the first unsupported measured native stage after write and source-item exposure have been established**. The positive forced side control further weakens a global-insensitivity account without changing b^* ; it does not identify a native latent attenuation onset.

This is still not an identified mediator coefficient. A causal mediation statement would additionally require compatible intervention surfaces and a noise/seed design that isolates each branch-specific atom. We do not have that evidence and do not make that claim.

2.5 PROSPECTIVE IMPLICATION: TEST SELECTIVE POST-EXPOSURE AUTHORIZATION

The completed experiment does not directly observe latent state relevance or latent “behavioral authority.” It establishes an operational boundary: source-item exposure can be frequent while a stable branch-conditioned first-action distribution shift remains unsupported by the frozen test. This leaves several explanations open, including content redundancy, policy priors, prompt/context attenuation, later-action effects, action-serialization ambiguity, and stochastic heterogeneity.

We therefore treat selective post-exposure authorization as a *prospective falsifiable hypothesis*, not as the mechanism already identified by C1. The clean future comparison asks whether a selector based on reproducible matched-state branch sensitivity outperforms an equally sparse outcome-blind random authorization rule. A stable branch contrast is not assumed to be correct, useful, or reward-improving; any future controller result must establish those properties separately. The full frozen PACTA-MSR protocol, including its matched-state reveal, developmental shadow gate, rate-matched random control, and unbiased exact-match-kernel MMD² endpoint, is retained in Appendix/Future Work so that a successor cannot move its goalposts. No PACTA-MSR method-effect result is reported in this paper.

2.6 INTERPRETATION DISCIPLINE

Three non-implications are load-bearing throughout the paper:

persistent-state divergence $\nRightarrow$ behavioral divergence,

retrieval exposure $\nRightarrow$ policy uptake,

forced leverage $\nRightarrow$ native end-to-end transport.

These distinctions also explain why our stopped evidence-gated residual extension is not a negative method result: that extension failed qualification before a behavioral method-effect experiment was authorized. The present paper therefore remains an identification/measurement study of the transport chain.

3 STAGE 1: REWARD-CONDITIONED WRITING ROBUSTLY CHANGES PERSISTENT STATE

We first test the write stage with paired interventions over released WebArena Shopping trajectories. Within each pair, the task text and full action history are fixed before the writer is called. One

branch uses the released success-reflection objective and the other uses the released failure-reflection objective; the resolved writer model is matched within pair and temperature is zero. Raw responses and the pre-writer trajectory projection are content-addressed before analysis.

The initial F0 tranche requested six source trajectories. Four returned complete memory outputs in both conditions; the two incomplete requests terminated in the failure-label writer before assistant text existed and were excluded from the paired content estimand. Among the four complete pairs, every pair changed normalized memory content and the extracted memory-item title set, with mean token-set Jaccard distance 0.735. Receipt-level forensics preserve the two incomplete arms as provider-response failures rather than treating them as negative memory examples.

The later frozen breadth tranche expands the write-stage evidence to 20 complete Shopping success/failure pairs. All 20/20 pairs produce different branch memories, with pooled token-set Jaccard distance 0.673. When combined with the four Reddit pairs used in the cross-domain replication, the D0 lineage contains 24/24 paired source experiences with the same pre-writer trajectory projection, the same resolved writer model within pair, temperature zero, and different branch-memory content. This establishes a robust operational branch contrast at the persistent-state level.

The content changes are not only token-level. A deterministic post-ready strategy-slot audit maps frozen memory titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification slots. In the original four complete F0 pairs, the strategy-slot set changes in all four. We treat this as a structural diagnostic rather than a semantic-validity claim: the current data do not certify each textual delta atom as a pure reward-caused proposition because explicit decoding seeds or same-condition noise-floor replications are unavailable.

The write-stage result therefore supports a narrow statement: under the released writer policy, flipping the reward-conditioned reflection branch on the same source experience reliably produces a different durable state. It does not by itself establish that the difference will be retrieved, followed, or reflected in terminal behavior. Section 4 first tests the strongest simple prompt-sensitivity reduction; Section 5 then measures downstream transport stage by stage.

4 RULING OUT GENERIC PROMPT SENSITIVITY

The success/failure writer-branch intervention changes the reflection instruction together with the outcome/reward semantics. It is therefore a bundled protocol contrast, not a pure reward-bit intervention. A generic prompt-sensitivity account predicts that a sufficiently strong wording perturbation inside one reflection mode should produce comparable memory divergence; the following control tests that reduction, but cannot decompose instruction semantics from reward semantics within the released branch. We evaluate it on eight released trajectories disjoint from F0 and balanced across original outcomes.

Before provider calls, we freeze semantic-preserving rewordings for both reflection modes. Under token-set Jaccard distance, the released success-versus-failure prompts differ by 0.114. The within-success rewording differs by 0.171 and the within-failure rewording by 0.152. The control thus perturbs surface wording more strongly while preserving the reflection objective.

Each fresh trajectory is written with the released success mode and the released failure mode. The same writer also processes one frozen semantic-preserving rewording of each mode at temperature zero. Let B_i be the memory distance between the two released reward modes and let W_i be the mean within-mode rewording distance. The frozen paired statistic is

$$\Delta_{\text{prompt}} = \frac{1}{8} \sum_{i=1}^8 (B_i - W_i). \quad (1)$$

All 32 control outputs complete. Mean reward-mode distance is 0.700, while the stronger same-mode wording control has mean 0.595. The paired excess is 0.105, and an exact one-sided sign-flip test gives $p = 0.0078$. This passes the frozen 0.10 effect margin together with the $p < 0.05$ gate. The reward-conditioned write effect therefore survives a same-information prompt-wording reduction that is deliberately stronger in lexical perturbation.

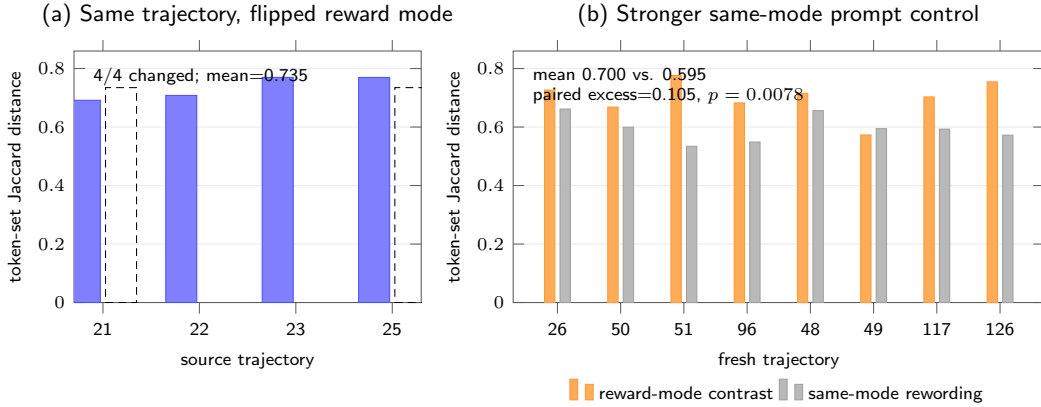


Figure 1: Write-channel evidence and its strongest simple control. (a) Among the four F0 pairs complete in both writer conditions, flipping only the reward-conditioned writer mode changes every memory. (b) On eight fresh control trajectories, reward-mode memory divergence exceeds stronger semantic-preserving same-mode prompt rewording. The control mean is 0.595 versus 0.700 between reward modes; paired excess is 0.105 with exact $p = 0.0078$.

Table 1: Evaluation-surface baselines on the same frozen system. Each coarsened evaluator supports a plausible but incomplete reading; the stage-resolved view is the minimum baseline that preserves where evidence is gained or lost.

Evaluator	Surface	Coarse reading	Alternative left aliased
Write-only	W	update is strong	persistent change does not establish downstream use
Retrieval-only	E	reuse is active	item availability is not decision-level uptake
Endpoint-only	O	native effect is near-null	writer, retrieval, and uptake failures collapse together
Forced-only	side F	memory has leverage	native retrieval/context transport is bypassed
Stage-resolved	W, E, U, O + side F	first unsupported measured native stage is localized	preserves stage identity instead of choosing one coarse projection

5 STAGE-RESOLVED TRANSPORT EVIDENCE

The write intervention becomes scientifically useful only once alternative explanations for weak downstream transport evidence are separated. We therefore use the frozen evidence in three roles: forced fixed-evidence exposure as a *capacity control*, native retrieval as an *availability measurement*, and first-action/terminal contrasts as *native transport measurements*. This section shows that their joint pattern is more informative than any endpoint alone; Table 1 makes the corresponding coarsened evaluation baselines explicit.

5.1 FORCED INJECTION ESTABLISHES DOWNSTREAM CAPACITY, NOT NATIVE TRANSPORT

We first bypass native retrieval and directly inject each paired memory into a fixed future-task packet. This asks whether the downstream system can respond to the paired persistent states when the experimenter supplies them.

The original fixed-state probe contains 12 aligned success-memory versus failure-memory rollouts; 7/12 choose different structured next actions. We then freeze all four complete initial F0 source-memory pairs and four future tasks before terminal-policy execution. Every source pair is crossed with every future task, giving 16 cells. The confirmatory run uses eight fresh rollouts per memory condition in every cell, for 256 completed calls. Mean absolute terminal success-rate difference is

0.15625, above the frozen 0.15 practical floor, and the 100,000-draw within-cell permutation test gives $p = 0.00074$.

This result weakens the explanation that the downstream system is simply incapable of responding to memory content. It does *not* estimate native transport because exposure is mechanically supplied. The cell effects are also heterogeneous: eight cells are zero, six have positive failure-minus-success effects, and two have negative effects; the two largest cells account for 78.2% of squared-effect mass. Even under forced exposure, a universal harmful direction is unsupported.

5.2 NATIVE SOURCE-ITEM EXPOSURE REMAINS COMMON WHILE ACTION UPTAKE IS UNSUPPORTED

We next restore the native memory-bank pathway. After expanding the frozen Shopping memory bank, the branch-conditioned source-memory item is retrieved on 125 of 172 held-out retrieval opportunities, a rate of 0.727. This item-level availability measurement does not identify exposure to the treatment-specific residual within the item, but it is sufficient to reject “the source memory item was never retrieved” as the sole description of the weak native downstream evidence. It also lets us test, rather than assume, the stronger evaluation shortcut “Retrieval hit is a surrogate for policy use.”

Despite that exposure, the first measured decision boundary is weak. Across the frozen 36 Shopping branch-comparison states, the original B10 probe used four repeated first-action decodes per success-memory branch and four per failure-memory branch. Its frozen primary mean total-variation distance is 0.06944 with $p = 0.5801$, and 0/36 states change their modal first action. A zero-provider provenance replay re-parses all 432 historical B10 raw responses under the frozen action normalizer and exactly reproduces the historical statistic. Because plug-in TV is upward-biased in sparse categorical samples, we additionally apply the exact-match-kernel unbiased collision MMD² statistic to the already-collected 4+4 branch draws. The post-hoc state-mean estimate is numerically zero (-6.2×10^{-18}), with one-sided state-stratified randomization $p = 0.5801$; 33/36 state estimates are zero, one is positive, and two are negative. A fixed-state descriptive bootstrap gives $[-0.0185, 0.0231]$. This reanalysis does not establish equivalence, but it shows that the original weak TV contrast is not strengthened after correcting for within-condition stochastic concentration. A separate post-hoc working-memory diagnostic yields branch shift 0.00335 with $p = 0.2052$. Native terminal transport is also small: mean absolute success difference is 0.02083 with permutation $p = 0.4289$, and 34/36 cells are exactly zero.

The conjunction is the stage-resolved evidence result. Applying the ordinal rule from Section 2.4, write is SUPPORTED, native source-item exposure is OBSERVED, and first-action uptake is the first measured native stage whose frozen primary test does not support a stable branch contrast. Thus $b^* = \text{first-action uptake}$ is an *evidence-localization boundary*—operationally, a boundary after exposure and before stable action uptake. Terminal sparsity is downstream corroboration rather than the quantity used to choose the boundary. Because the stage tests differ in units and statistical sensitivity, this does not identify where a latent causal effect physically attenuates; in particular, it is not a causal mediation coefficient.

A structural sensitivity audit clarifies what is load-bearing in this localization. Removing terminal outcome evidence, removing the Reddit terminal tranche, or removing the forced-capacity side control leaves the first unsupported measured native stage at first-action uptake. In contrast, merging native exposure with first-action uptake removes the diagnostic resolution needed to distinguish availability from behavioral use. The contribution is therefore not simply to report more stage metrics: for this frozen system, the *separation of exposure from uptake* is the load-bearing measurement that prevents these two explanations from becoming observationally aliased.

The full six-hypothesis alternative-explanation audit is reported in Appendix Table 6; the main-text stage ladder retains the decision-critical evidence states without spending a second full table on the same inference chain.

5.3 CROSS-DOMAIN BOUNDARY: WRITE REPLICATION WITHOUT MONOTONE TRANSPORT

We repeat the write-versus-native-transport measurement on Reddit. Reward-conditioned writing again diverges in 4/4 paired source memories. Native terminal transport is larger than Shopping in magnitude but remains unresolved and highly sparse: mean absolute success difference is 0.125 with

Table 2: Stage-evidence ladder. Evidence states preserve the semantics of each measurement instead of forcing heterogeneous quantities onto one scale. The forced arm is a side capacity control and therefore does not enter the native-stage ordering.

Stage	Evidence state	Frozen observation / interpretation
Persistent write	Supported	20/20 Shopping pairs diverge; stronger same-mode control leaves excess 0.105, exact $p = 0.0078$.
Forced capacity (side control)	Supported	$ \Delta = 0.15625$, $p = 0.00074$; supplied memory has downstream leverage but native retrieval is bypassed.
Native source-item exposure	Directly observed	125/172 held-out opportunities retrieve the branch-conditioned source-memory item; this is item availability, not treatment-residual exposure or policy use.
First-action uptake	Not supported	Primary TV = 0.06944, $p = 0.5801$, 0/36 modal changes; replay-qualified post-hoc collision $\widehat{\text{MMD}}_u^2 \approx 0$, $p = 0.5801$.
Native outcome	Sparse / heterogeneous	Shopping 34/36 zero; Reddit 6/8 zero and the two nonzero signs are opposite.

PERSISTENT WRITE **NATIVE EXPOSURE** **FIRST-ACTION UPTAKE** **NATIVE OUTCOME**

20/20 Shopping pairs diverge
Jaccard = 0.673

125/172 retrieval hits
rate = 0.727

TV = 0.06944, $p = 0.5801$
0/36 modal changes

Shopping $|\Delta| = 0.02083$
 $p = 0.4289$; 34/36 zero

strongest supported attenuation boundary: after exposure, before stable action uptake
operational localization only - not a causal mediation coefficient

FORCED CAPACITY CONTROL
fixed-evidence terminal $|\Delta| = 0.15625$
 $p = 0.00074$

**Cross-domain boundary: Reddit write 4/4 diverges; native terminal $|\Delta| = 0.125$, $p = 0.2253$;
6/8 cells are zero and the two nonzero cells have opposite signs.**

Figure 2: **Native transport and forced capacity are different experimental surfaces.** The horizontal path is the deployed reuse chain: persistent write $\rightarrow$ native source-item exposure $\rightarrow$ first-action uptake $\rightarrow$ native outcome. The forced fixed-evidence arm bypasses retrieval and is shown as a side capacity control. On the frozen Shopping tests, first-action uptake is the first unsupported measured native stage after write and item exposure are established; this is an evidence boundary, not a latent causal bottleneck or mediation coefficient.

$p = 0.2253$, 6/8 cells are zero, and the two nonzero cells have opposite signs. The second domain therefore supports the same coarse structural separation—robust write divergence need not become stable native terminal transport—while rejecting a simple universal attenuation constant or signed harm coefficient. Reddit does not contain the matched native exposure and first-action measurements needed to replicate the Shopping evidence boundary.

An outcome-blind structured-memory control gives a complementary warning about endpoint interpretation. It produces a small terminal contrast of 0.04514 ($p = 0.0048$), below the frozen 0.15 practical floor, with 32/36 cells zero. Statistical detectability alone is therefore not the scientific target; a sparse small effect can pass a null test without matching the magnitude or mechanism of forced branch leverage.

5.4 WHY THE ONE-STEP REWARD-TO-BEHAVIOR STORY FAILS

A one-step interpretation would take the 256 forced rollouts as evidence that “reward error propagates to future behavior.” The native measurements show why this is too coarse. Forced injection estimates leverage conditional on making a particular memory state available; native transport additionally

depends on retrieval competition, context composition, and policy uptake. Indeed, coarsened views of the same frozen system disagree: write-only looks strongly affected, retrieval-only looks active, native endpoint-only looks nearly null, and forced-only shows leverage. The full stage signature reconciles these summaries rather than averaging them. The difference between 0.15625 forced leverage and 0.02083 native Shopping terminal transport is therefore not a scalar attenuation factor to divide or average; the two surfaces answer different questions.

Conversely, looking only at the native endpoint would also lose the mechanism: a 0.02083 Shopping terminal difference does not tell us that the writer failed, that memory was unavailable, or that the downstream system was globally insensitive. The stage-resolved controls exclude or weaken each of those simpler accounts before the first-action boundary is reached.

The evaluation consequence is concrete. For a persistent update, report at least one state-change metric, one exposure measurement, and one downstream uptake/outcome metric; use a capacity control only if its bypassed stages are explicit. Then report the first measured stage where evidence for transport is no longer supported, while keeping stage-specific power and measurement semantics explicit, rather than crediting the update with behavioral authority because its representation changed or because a retrieved item appeared in context.

6 RELATED WORK

Feedback-conditioned memory. Prior work covers reward reliability (Wang et al., 2026), success/failure-derived memory (Ouyang et al., 2026), reward-conditioned construction (Wang et al., 2025), and scoring/reuse distortion (Asadolahi et al., 2026; Yang et al., 2026). These are not our novelty; the frozen writer branch is only our controlled intervention.

Retrieval, utilization, and our residual. Mem2ActBench studies active use (Shen et al., 2026); QCR separates retrieval from reuse (Li et al., 2026); Yuan et al. diagnose write/retrieval/utilization (Yuan et al., 2026); and lifecycle work names the same broad stages (Lin et al., 2026). These concepts are not our novelty. We instead hold the source trajectory fixed and trace the bundled writer-protocol contrast through write, native exposure, first-action uptake, outcome, and forced capacity. The load-bearing distinction is separation of exposure from uptake, not lifecycle naming or a power-matched causal bottleneck; known self-improvement variance provides complementary motivation (Ye et al., 2026).

7 LIMITATIONS

The released success/failure branch jointly changes reflection instruction and outcome/reward semantics; per-atom decoding-seed binding at the writer intervention is unavailable, two failure-label calls ended before assistant text, and source-faithful live WebArena navigation is unavailable. We therefore estimate a bundled writer-protocol contrast, not a pure reward-bit effect. The first-action B10 audit does contain repeated same-condition decodes and now has a replay-qualified collision/MMD² reanalysis, but that reanalysis is post-hoc rather than an independent prospective replication and therefore does not license an equivalence claim.

Stage tests differ in units, samples, power, and intervention surfaces, so first-action uptake is only an *evidence-localization boundary*, not an identified latent attenuation onset or mediator. The 125/172 statistic is source-item retrieval, not residual exposure or policy use; forced injection bypasses native retrieval/context composition and measures supplied-evidence capacity.

Shopping is sparse (34/36 zero); Reddit is sparse, sign-heterogeneous (6/8 zero), and lacks matched exposure/action measurements, so neither a universal effect nor cross-domain action boundary is claimed. External validity is restricted to the frozen released writer/policy configuration and the Shopping+Reddit substrate evaluated here; other writers, backbones, domains, and memory architectures are untested. The CBRG extension stopped before behavioral evaluation; no behavioral repair efficacy is claimed.

8 CONCLUSION

Coarse views of the same update disagree. In Shopping, first-action uptake is the first unsupported measured native stage after write and source-item exposure—an evidence boundary, not a latent causal bottleneck. Thus **state divergence and retrieval exposure do not by themselves establish stable decision-level uptake**. The current contribution is stage-resolved identification of that boundary; selective post-exposure authorization remains a separate prospective hypothesis.

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A F0 REPRODUCTION DETAILS

A.1 RELEASED TRAJECTORY SAMPLE

The F0 sample uses six task IDs from the released AWM shuffle-seed-42 Shopping parquet: 21 through 25, plus 47. The deterministic sampler selects the lowest task IDs with parseable trajectory records within each original-outcome stratum. Original failures are 21 through 22 plus 24. Original successes are 23 plus 25 and 47.

A.2 MEMORY CONSTRUCTION INTERVENTION

Each released trajectory is summarized into an action trace. The same trace is inserted into two prompts. One prompt contains the released ReasoningBank success-reflection system instruction. The paired prompt contains the released failure-reflection system instruction. The task prompt stays fixed. The writer model stays fixed. Temperature stays at zero. The raw provider output is archived before metric computation.

Table 3: Complete paired interventions.

Task	Original outcome	Content changed	Token distance
21	failure	yes	0.691489
22	failure	yes	0.708108
23	success	yes	0.769953
25	success	yes	0.769608

A.3 PAIR-LEVEL F0 RESULTS

The extracted memory-item title set changes for every complete pair. The mean token-set Jaccard distance is 0.734789. A deterministic post-ready strategy-slot audit maps the frozen titles into navigation, pagination/coverage, query expansion, semantic matching, evidence capture, extraction, and verification/completeness. All four complete pairs change their slot set. The slot-set Jaccard distances are 0.571, 0.400, 0.500, and 0.500 for tasks 21, 22, 23, and 25, respectively, with mean 0.493. This is a structural diagnostic over frozen titles, not an embedding-based semantic claim.

Provider failures for task 24 and task 47 are excluded from the paired analysis. Both are failure-label calls with `ArkResponseStateError`; neither produces assistant text or a raw memory SHA. Private response identifiers are retained and GET-only recovery finds no text. The four completed failure-label calls produce 777, 852, 1,113, and 1,296 output tokens. These receipts distinguish provider response-state failures from a later memory-parser rejection, while leaving arm-dependent completion bias unresolved.

B PROMPT-PERTURBATION CONTROL DETAILS

The control set is disjoint from the F0 requests. Each fresh trajectory is written under both released reward modes. One semantic-preserving rewording is also evaluated within each mode. Table 4 reports the paired distances used in the frozen sign-flip test.

Table 4: Full fresh prompt-perturbation control. Between is the released success-versus-failure memory distance; Within averages the two same-mode rewording distances.

Task	Original outcome	Between	Within	$B_i - W_i$
26	failure	0.726	0.662	0.065
50	failure	0.668	0.599	0.069
51	failure	0.776	0.534	0.242
96	failure	0.682	0.549	0.134
48	success	0.716	0.656	0.059
49	success	0.573	0.594	-0.022
117	success	0.703	0.593	0.110
126	success	0.755	0.572	0.182
Mean		0.700	0.595	0.105

C DOWNSTREAM CONFIRMATORY DETAILS

C.1 FIXED-STATE ACTION DISTRIBUTIONS

The distributional action stress test retains all four intervention states from the action-reach experiment and uses eight rollouts per memory condition. The per-state success-memory versus failure-memory TV distances are 0.625 for task 22 and 0.375 for task 23. They are 0.125 for task 24 and 0.250 for task 26. Their mean is 0.34375. The frozen 100,000-draw permutation audit gives $p = 0.310527$.

C.2 FROZEN TERMINAL MATRIX

The terminal confirmatory experiment reuses every complete F0 pair and every future task frozen for the earlier fixed-evidence run. Source-memory tasks form $\{21, 22, 23, 25\}$. Future tasks form $\{164, 385, 387, 388\}$. The design therefore contains 16 source-future cells. Each cell receives eight

fresh success-memory rollouts and eight fresh failure-memory rollouts. The policy model is Doubao Seed 2.0 Mini at temperature 0.2. All 256 requested calls complete with zero provider failures.

The primary statistic is the mean across cells of the absolute difference between success-memory and failure-memory terminal benchmark success rates. The support gate is frozen before calls at an effect floor of 0.15 together with a within-cell permutation $p < 0.05$. The permutation audit uses 100,000 draws with seed 20260822. The observed statistic is 0.15625 and the audit gives $p = 0.00074$.

Table 5: F2R1 confirmatory terminal success rates. S denotes success-label memory and F denotes failure-label memory.

Source	Future	S rate	F rate	$ \Delta $
21	164	1.000	1.000	0.000
21	385	0.000	0.250	0.250
21	387	0.125	0.250	0.125
21	388	1.000	0.625	0.375
22	164	1.000	1.000	0.000
22	385	0.000	0.125	0.125
22	387	0.125	0.000	0.125
22	388	0.250	1.000	0.750
23	164	1.000	1.000	0.000
23	385	0.000	0.000	0.000
23	387	0.125	0.250	0.125
23	388	1.000	1.000	0.000
25	164	1.000	1.000	0.000
25	385	0.000	0.000	0.000
25	387	0.125	0.750	0.625
25	388	1.000	1.000	0.000
Mean absolute difference				0.15625

C.3 HETEROGENEITY AND CORRUPTION-RATE DECOMPOSITION

For each frozen cell, the confirmatory run estimates paired conditional success probabilities. Eight of the 16 cells have zero observed success-rate contrast. Six have positive failure-memory minus success-memory effects and two have negative effects. The largest squared effects occur at source/future pairs 22/388 and 25/387; together they contribute 78.2051% of the total squared-effect mass. This sign and magnitude heterogeneity is why the revised paper treats the experiment as a forced-leverage measurement rather than a uniform directional effect or a native end-to-end transport estimate.

If reward-label corruption selects the failure-conditioned memory with probability q , the induced between-memory variance equals $q(1 - q)\Delta_c^2$ in cell c . The exact mean squared conditional effect across all 16 cells is 0.076172. Averaging over cells gives the curve $0.076172q(1 - q)$, with value 0.006855 at $q = 0.1$, value 0.014282 at $q = 0.25$, and value 0.019043 at $q = 0.5$. This remains a descriptive plug-in calculation conditional on forced memory exposure; it is not used as the main claim in the revised stage-resolved paper.

C.4 NATIVE TRANSPORT AUDIT ADDED AFTER THE FORCED-LEVERAGE STUDY

The later frozen native-pipeline audit changes the interpretation of the forced matrix. In Shopping, the branch-conditioned source-memory item is retrieved in 125/172 held-out opportunities after bank expansion, but the mean absolute native terminal success difference is only 0.02083 with permutation $p = 0.4289$; 34/36 terminal cells are zero. The corresponding first-action total-variation statistic is 0.06944 with $p = 0.5801$, and no frozen state changes its modal first action between reward branches. A post-hoc working-memory branch-shift diagnostic is 0.00335 with $p = 0.2052$.

Replay-qualified first-action stochasticity diagnostic. Historical B10 already contains four success-memory, four failure-memory, and four no-memory first-action draws for each of the 36 matched Shopping states. A zero-provider provenance qualification recovers all 432 stage records, content-addressed raw responses, and provider receipts; replaying each raw response through the frozen historical parser reproduces all 432 stored action signatures and exactly recovers the original mean TV 0.0694444444, 100,000-draw state-stratified permutation $p = 0.5800941991$, and 0/36 modal changes.

To avoid interpreting sparse plug-in categorical TV as a stochasticity-adjusted distance, we reuse only the historical success/failure 4+4 draws and apply the exact-match kernel $k(a, b) = \mathbf{1}[a = b]$. For state i , the unbiased two-sample estimator is

$$\widehat{\text{MMD}}_{u,i}^2 = \frac{1}{m(m-1)} \sum_{j \neq \ell} k(s_j, s_\ell) + \frac{1}{n(n-1)} \sum_{j \neq \ell} k(f_j, f_\ell) - \frac{2}{mn} \sum_{j, \ell} k(s_j, f_\ell),$$

with $m = n = 4$. Finite-sample negative values are retained. The panel mean is numerically zero (-6.17×10^{-18}); 33/36 state estimates are zero, one is $+1/3$, and two are $-1/6$ each. The pre-outcome-frozen 100,000-draw within-state branch-label randomization gives one-sided $p = 0.5800941991$, and a fixed-state descriptive bootstrap gives a 95% interval $[-0.018519, 0.023148]$. This is a post-hoc diagnostic on prospectively collected historical draws, not an independent prospective replication. It supports the conservative statement that stable first-action branch separation is not established after accounting for within-condition stochastic concentration; it does not establish equality or a zero effect.

The cross-domain Reddit audit contains four complete success/failure writes, all of which diverge. Its native terminal mean absolute difference is 0.125 with $p = 0.2253$; 6/8 cells are zero and the two nonzero cells have opposite signs. These later measurements are why the main text separates forced capacity from native transport and treats first-action uptake as the first unsupported measured Shopping stage after write and source-item exposure, rather than as an identified latent attenuation onset or causal mediation result.

For finite-cell sensitivity, we run a fixed-seed nonparametric bootstrap with 100,000 repetitions, resampling 16 source-future cells with replacement. The percentile 95% interval for mean absolute effect is $[0.054688, 0.273438]$. The corresponding interval for mean squared effect is $[0.010742, 0.164062]$, which maps to $[0.002686, 0.041016]$ for $V(0.5)$. This bootstrap is descriptive sensitivity to the observed cell composition. The preregistered within-cell permutation test on the 256 confirmatory rollouts supplies the primary significance result.

C.5 FULL ALTERNATIVE-EXPLANATION AUDIT

Table 6: Alternative-explanation audit. “Weakened” means the frozen evidence makes the explanation incomplete but does not mathematically eliminate every variant of it.

Explanation	Frozen evidence	Disposition
No persistent state intervention	20/20 Shopping and 4/4 Reddit writes diverge; same-mode excess = 0.105, $p = 0.0078$	Inconsistent with the observed write-stage intervention.
Downstream policy is globally insensitive to memory	Forced terminal $ \Delta = 0.15625$, $p = 0.00074$	Weakened: supplied branch-specific memory has measurable leverage.
Native null is only retrieval absence	125/172 native Shopping source-item retrieval hits	Weakened: substantial item exposure coexists with weak first-action contrast.
Retrieval hit is a surrogate for policy use	first-action TV = 0.06944, $p = 0.5801$; replay-qualified post-hoc $\widehat{\text{MMD}}_u^2 \approx 0$, $p = 0.5801$	Rejected as an evaluation equivalence: item availability is not policy uptake.
Universal directional terminal transport	Shopping 34/36 zero; Reddit 6/8 zero, two nonzero signs opposite	Not supported by the frozen native cells.
Causal mediation has been identified	bundled writer intervention lacks per-atom seed binding; collision audit is post-hoc; forced/native surfaces differ	Unresolved and explicitly outside the claim set.

D PROSPECTIVE SELECTIVE-AUTHORIZATION HYPOTHESIS

The completed C1 evidence does not identify a latent “behavioral authority” variable and does not establish that selective authorization is the missing mechanism. It leaves a narrower falsifiable successor question: after memory content is already available, can a selector based on reproducible matched-state branch sensitivity outperform an equally sparse outcome-blind random authorization rule?

We freeze the prospective PACTA-MSR comparison only to prevent post-outcome goalpost movement. The same source trajectory produces paired success/failure writer states. A deterministic branch-blind matched-state reveal supplies identical current-state evidence before authorization. A developmental shadow statistic, $\gamma_i = \min(B_{1i}, B_{2i}) - \max(W_{Si}, W_{Fi})$, is used only for qualification/localization. In the conditional final comparison, A_3 authorizes units selected by the frozen gate and A_2 authorizes the same number using a frozen outcome-blind random ranking. The final branch-separation statistic is the unbiased exact-match-kernel MMD^2 /collision U-statistic, with $D_{\text{select}} = U_{A_3} - U_{A_2}$; finite-sample negative values are retained and raw empirical TV is descriptive only. The frozen pilot gate uses mean $D_{\text{select}} \geq 0.20$, more positive than negative unit-level contrasts, mean $(A_3 - A_0) > 0$, and mean $(A_3 - A_1) \geq 0$ on the same MMD^2 scale.

These definitions specify a future experiment, not a present method result. A stable branch contrast is not assumed to be correct, useful, or reward-improving. Any future method claim additionally requires canonical action identity, disjoint shadow/final samples, nondegenerate opening count, and uncertainty at the source/future-unit level. No current C1 conclusion depends on PACTA-MSR succeeding.